BLDC-ESC / v1 / schematic draft

01_dc_link

XT60 / DC-link / TVS / MT60

File: schematic/01_dc_link.kicad_sch

02_bridge

MOSFETs / shunts / INA241

File: schematic/02_bridge.kicad_sch

03_regen

Bus clamp / dump switch

File: schematic/03_regen.kicad_sch

04_power_usb

Buck / power mux / 3V3

File: schematic/04_power_usb.kicad_sch

05_mcu

STM32G474 / reference / SWD

File: schematic/05_mcu.kicad_sch

06_sensors

Hall / encoder / sensor power

File: schematic/06_sensors.kicad_sch

07_io_can

CAN FD / UART

File: schematic/07_io_can.kicad_sch

08_analog_inputs

Throttle / RC PWM / digital I/O

File: schematic/08_analog_inputs.kicad_sch

09_safety

Arm latch / PWM buffer / interlocks

File: schematic/09_safety.kicad_sch

10_voltage_sensing

Bus and phase ADC buffers

File: schematic/10_voltage_sensing.kicad_sch

11_current_protection

Hardware current and bus windows

File: schematic/11_current_protection.kicad_sch

12_gate_driver

DRV8353 / gate supplies / SPI

File: schematic/12_gate_driver.kicad_sch

13_usb_status

USB-C / ESD / status LEDs

File: schematic/13_usb_status.kicad_sch

14_temperature

NTCs / analog rail monitor

File: schematic/14_temperature.kicad_sch

15_regen_safety

Chopper OCP / latch / telemetry

File: schematic/15_regen_safety.kicad_sch

16_supervisor

Reset / watchdog / service mode

File: schematic/16_supervisor.kicad_sch

X = DNP / not fitted. v1 draft; electrical review and bench validation pending.

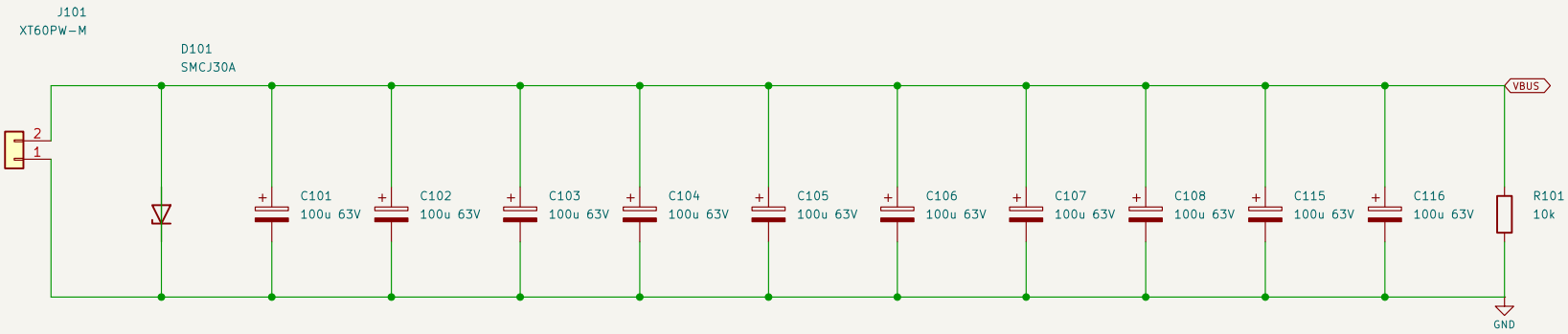
Sheet: /
File: bldc-esc.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 1/17

DC input and motor connections

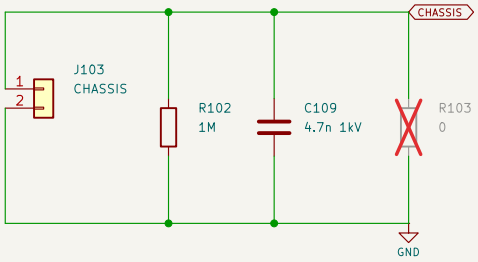
DC input and local energy storage



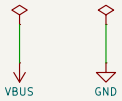
12–30 V operating bus. External DC-rated fuse required close to the source. No electronic reverse-polarity protection.

C101–C108, C115–C116: 10 x 100 uF / 63 V hybrid, 1,000 uF total. Bulk bank does not absorb sustained regeneration.

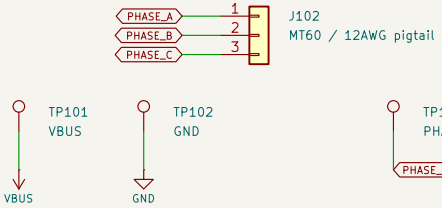
Shield / chassis connection



R103 is DNP. Select the chassis bond after enclosure and EMI review.



Motor cable / test points

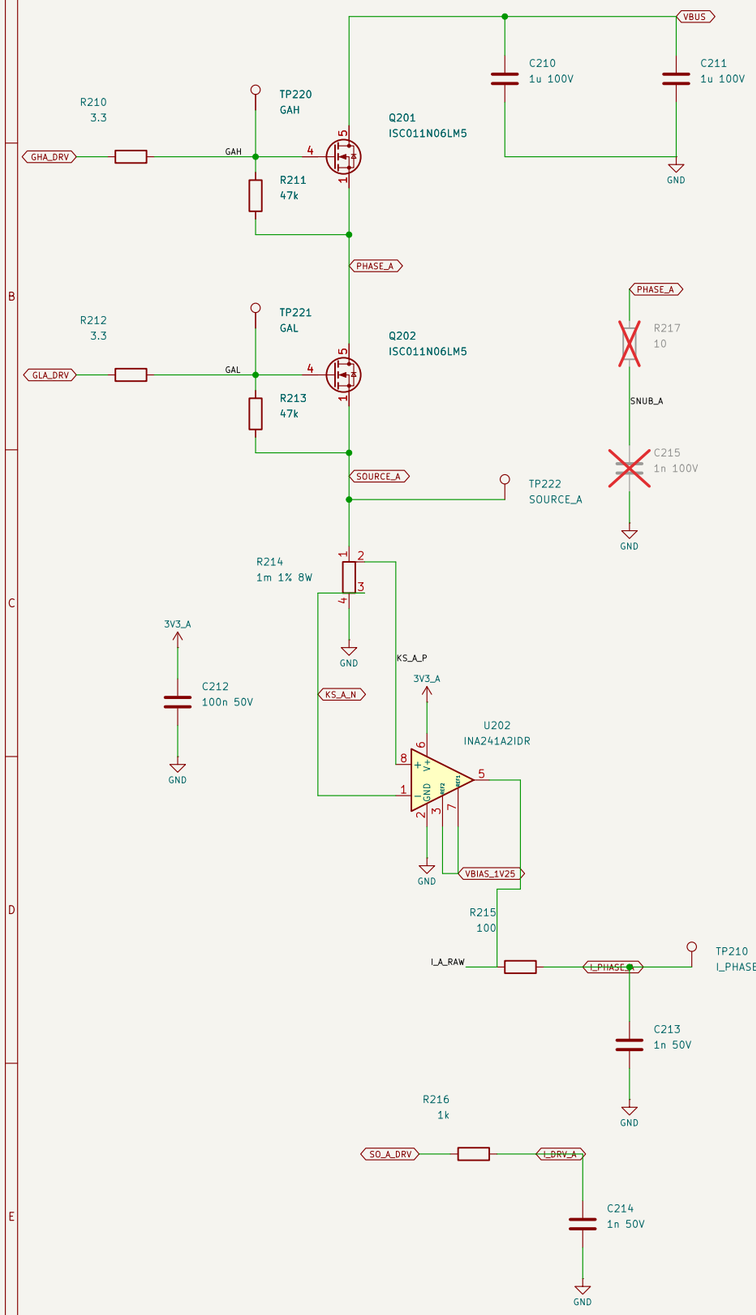


MT60 pigtail: 1=A, 2=B, 3=C. Provide strain relief.

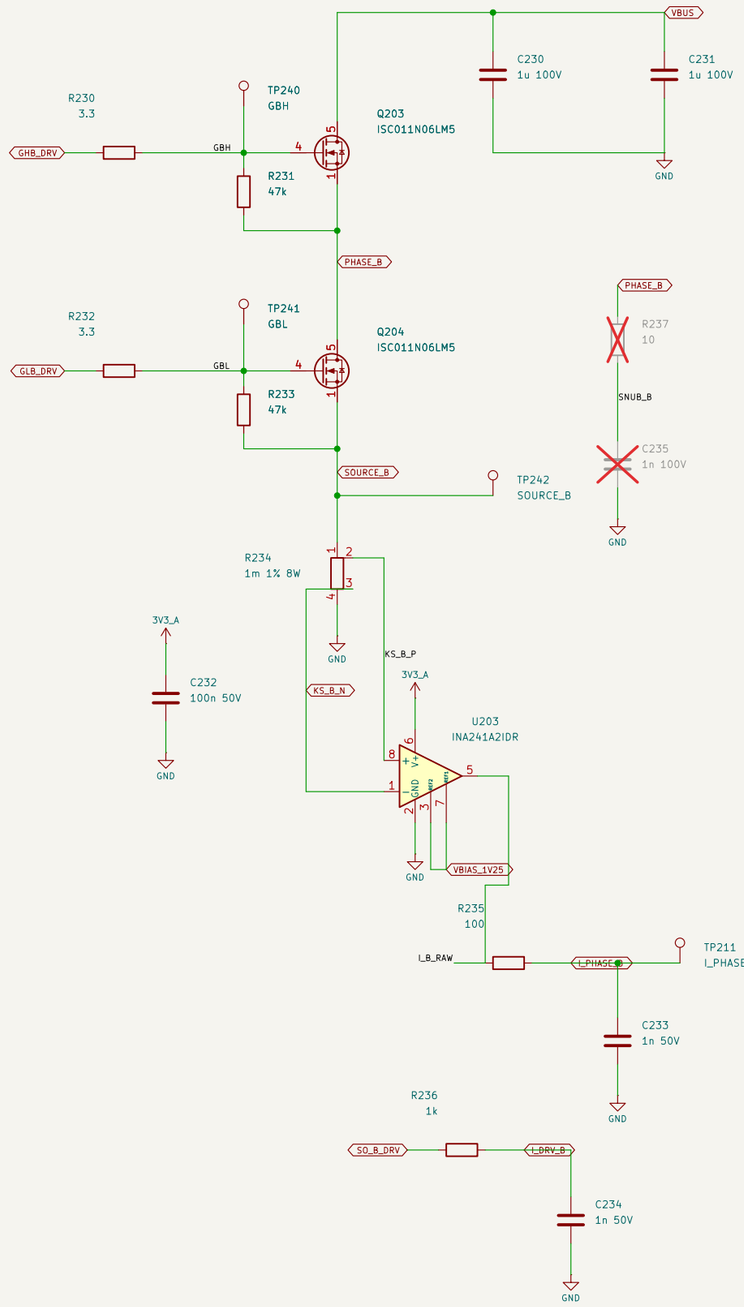
Sheet: /01_dc_link/	
File: 01_dc_link.kicad_sch	
Title: DC input and motor connections	
Size: A4	Date:
KiCad E.D.A. 10.0.4	**Rev: v1-draft**
	Id: 2/17

Three-phase bridge and current sensing

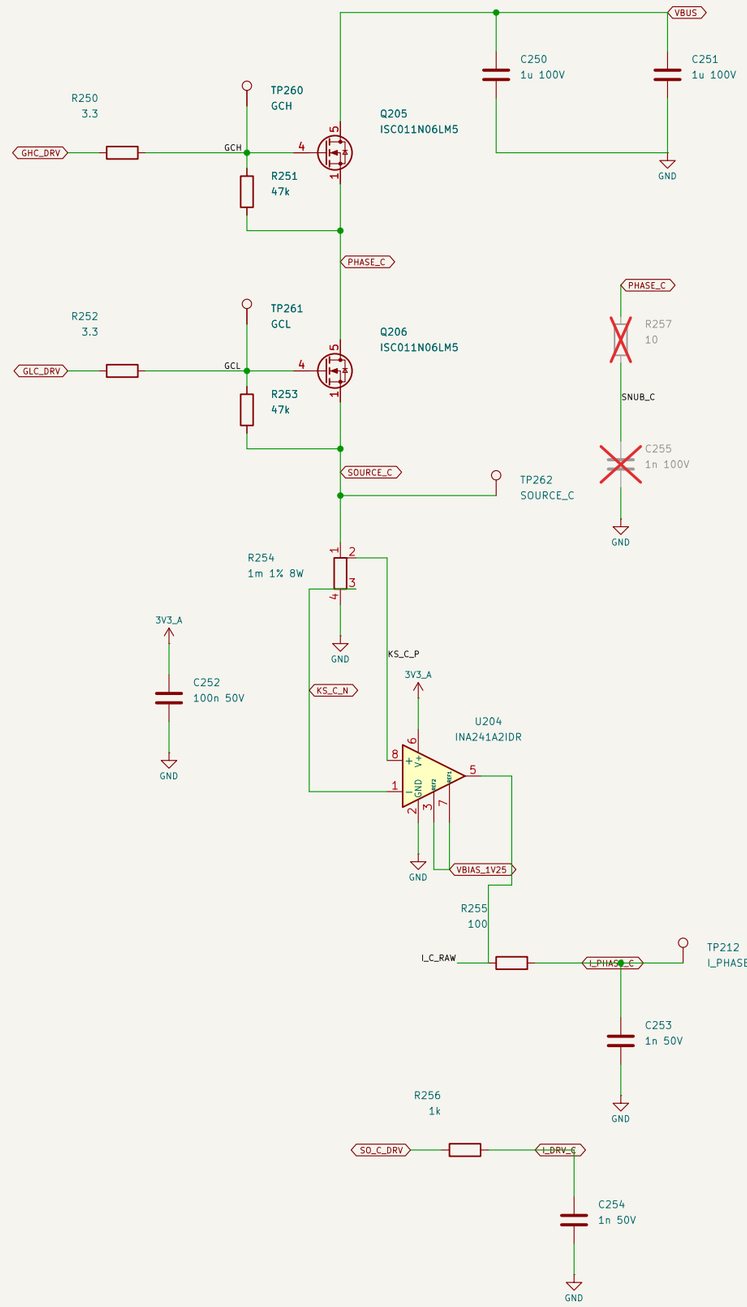
Phase A



Phase B



Phase C

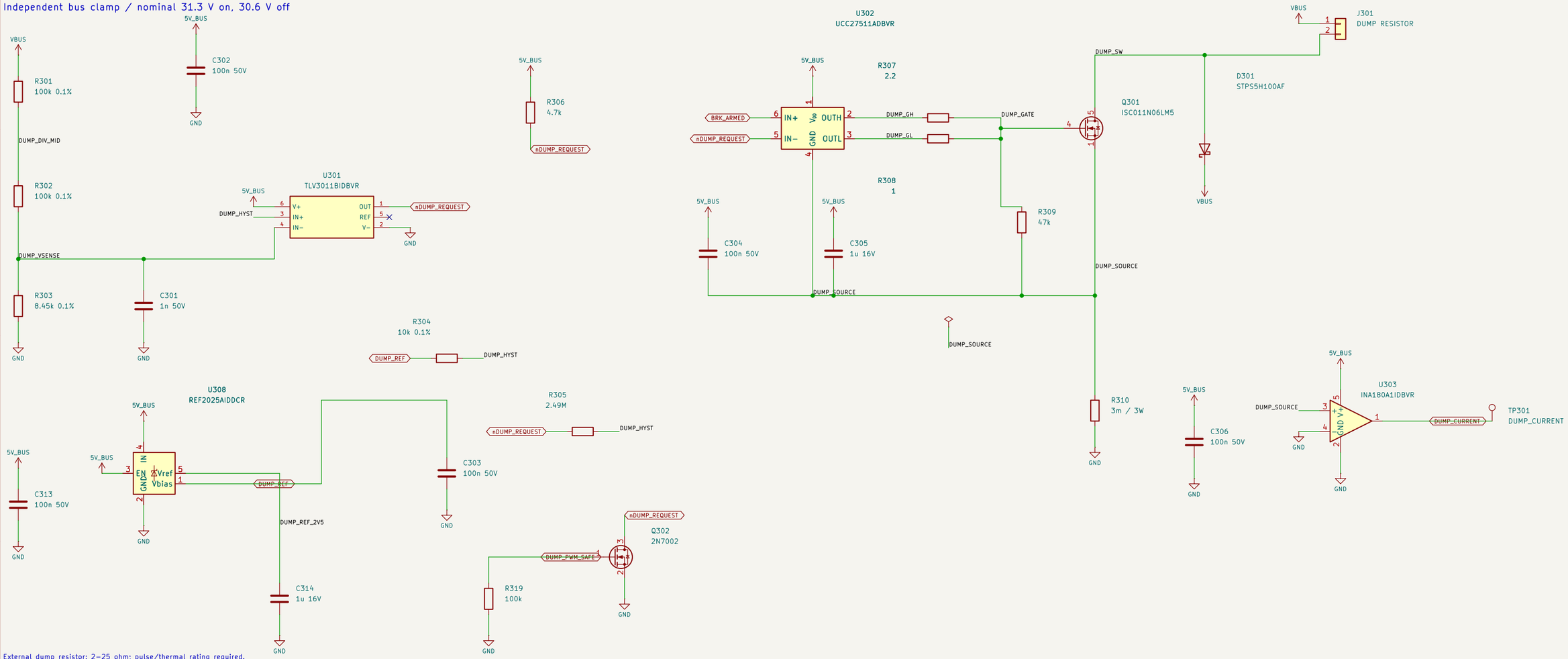


Main CSA: 1.25 V zero, 20 mV/A
Gate probes need local source returns.
High-side VGS: differential probe only.

Sheet: /02_bridge/		
File: 02_bridge.kicad_sch		
Title: Three-phase bridge and current sensing		
Size: A3	Date:	**Rev: v1-draft**
KiCad E.D.A. 10.0.4		Id: 3/17

Regenerative brake chopper

Independent bus clamp / nominal 31.3 V on, 30.6 V off



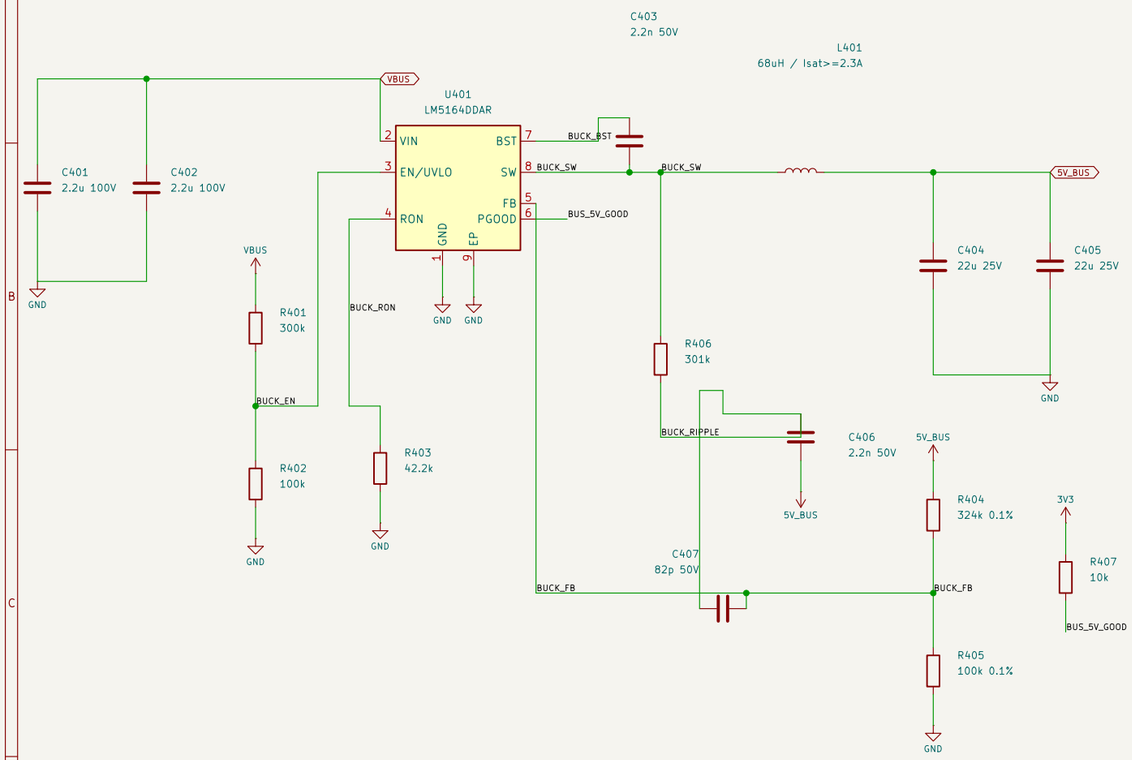
External dump resistor: 2–25 ohm; pulse/thermal rating required.
Chopper trip: about 20.8 A, power-cycle reset. Reference U308 stays bus-powered.

U302 return is Kelvin-connected to the top of R310.
Do not force a bench supply to sink regenerative current.

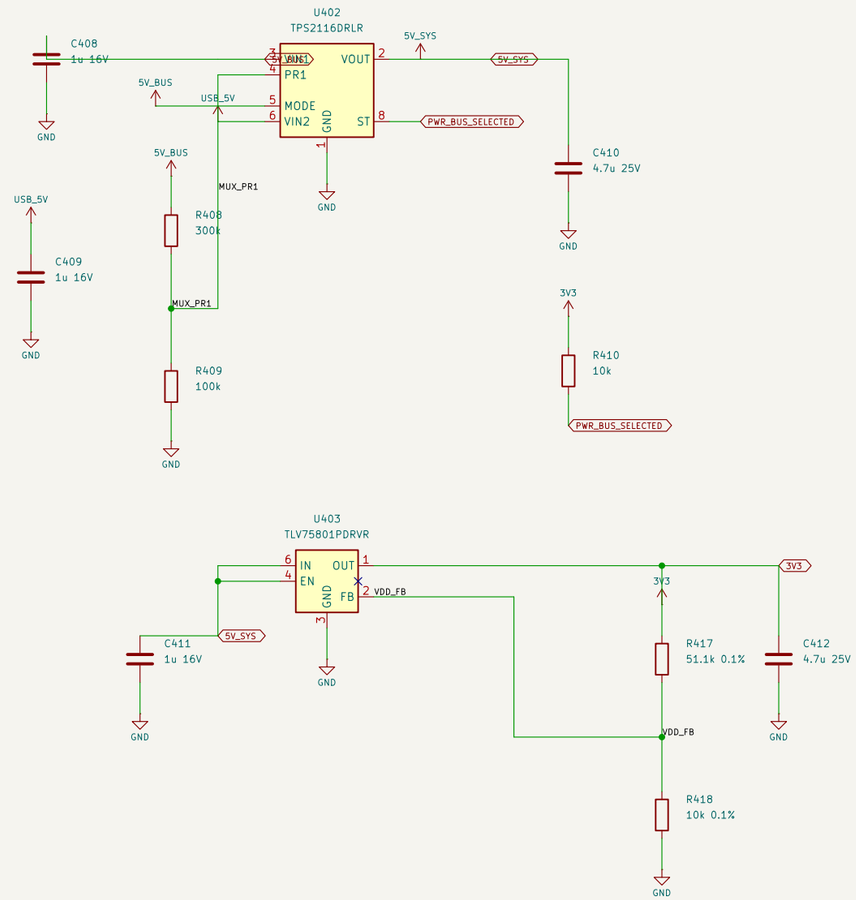
Sheet: /03_regen/	
File: 03_regen.kicad_sch	
Title: Regenerative brake chopper	
Size: A3	Date:
KiCad E.D.A. 10.0.4	Id: 4/17
	Rev: v1-draft

Power supplies

Bus supply / LM5164, nominal 5 V



Bus / USB selection and logic supply



68 uH / about 300 kHz. R406, C406, C407 provide type-3 ripple injection. Verify startup and load transients.

3V3 net: 3.3605 V nominal (51.1k / 10k).
USB-only: bridge disarmed; sensor supplies off at boot.

Sheet: /04_power_usb/		
File: 04_power_usb.kicad_sch		
Title: Power supplies		
Size: A3	Date:	**Rev: v1-draft**
KiCad E.D.A. 10.0.4		Id: 5/17

MCU, reference and debug

STM32G474VET6 / TIM1 bridge control

U501
STM32G474VET6

Y501
8MHz / CL18p

C513
27p 50V

C514
27p 50V

R502
0

8 MHz crystal: load capacitors
remain subject to startup validation.

R504
10k

SW501
RESET

C515
100n 50V

VDDA and reference

FB501
600R @ 100MHz

C507
4.7u 25V

C508
100n 50V

D501
PMEG2010ER

C509
100n 50V

U502
REF2025AIDDCR

C510
1u 16V

C511
100n 50V

R501
1k

C512
100n 50V

R503
100k

PB8 is BOOT0. PA15 carries I2C1 SCL.

Programming / debug

J501
SWD / KEY 7

R505
10k

SW502
USER

TP501
VREF_2V5

TP502
VBIAS_1V25

TP503
3V3_A

TP504
ADC_TRIG_SCOPE

TP505
DAC1_CH1_TP

TP506
DAC1_CH2_TP

Nonvolatile configuration

R506
4.7k

R507
4.7k

U503
M24C64-RMN6TP

R508
10k

C516
100n 50V

MCU bypass capacitors

C501
100n 50V

C502
100n 50V

C503
100n 50V

C504
100n 50V

C505
100n 50V

C517
100n 50V

C506
4.7u 25V

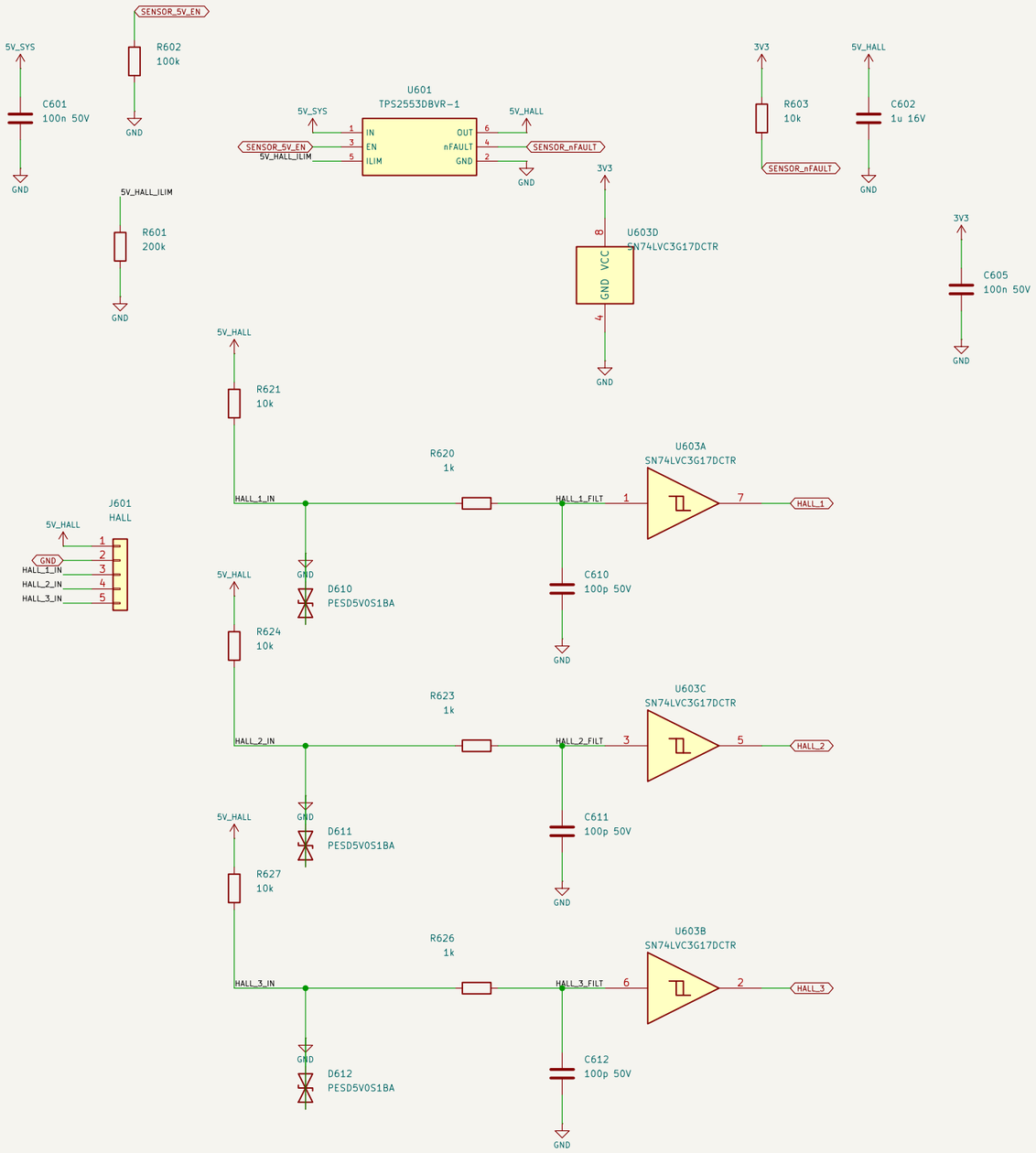
VREF+ = 2.5 V; current midpoint = 1.25 V.
Reference supply tracks VDDA during brownout.

Injected ADC conversions use TIM1 TRGO. Configure break inputs, deadtime and safe GPIO states before requesting bridge arm.

Sheet: /05_mcu/	
File: 05_mcu.kicad_sch	
Title: MCU, reference and debug	
Size: A3	Date:
KiCad E.D.A. 10.0.4	**Rev: v1-draft**
	Id: 6/17

Hall and encoder interfaces

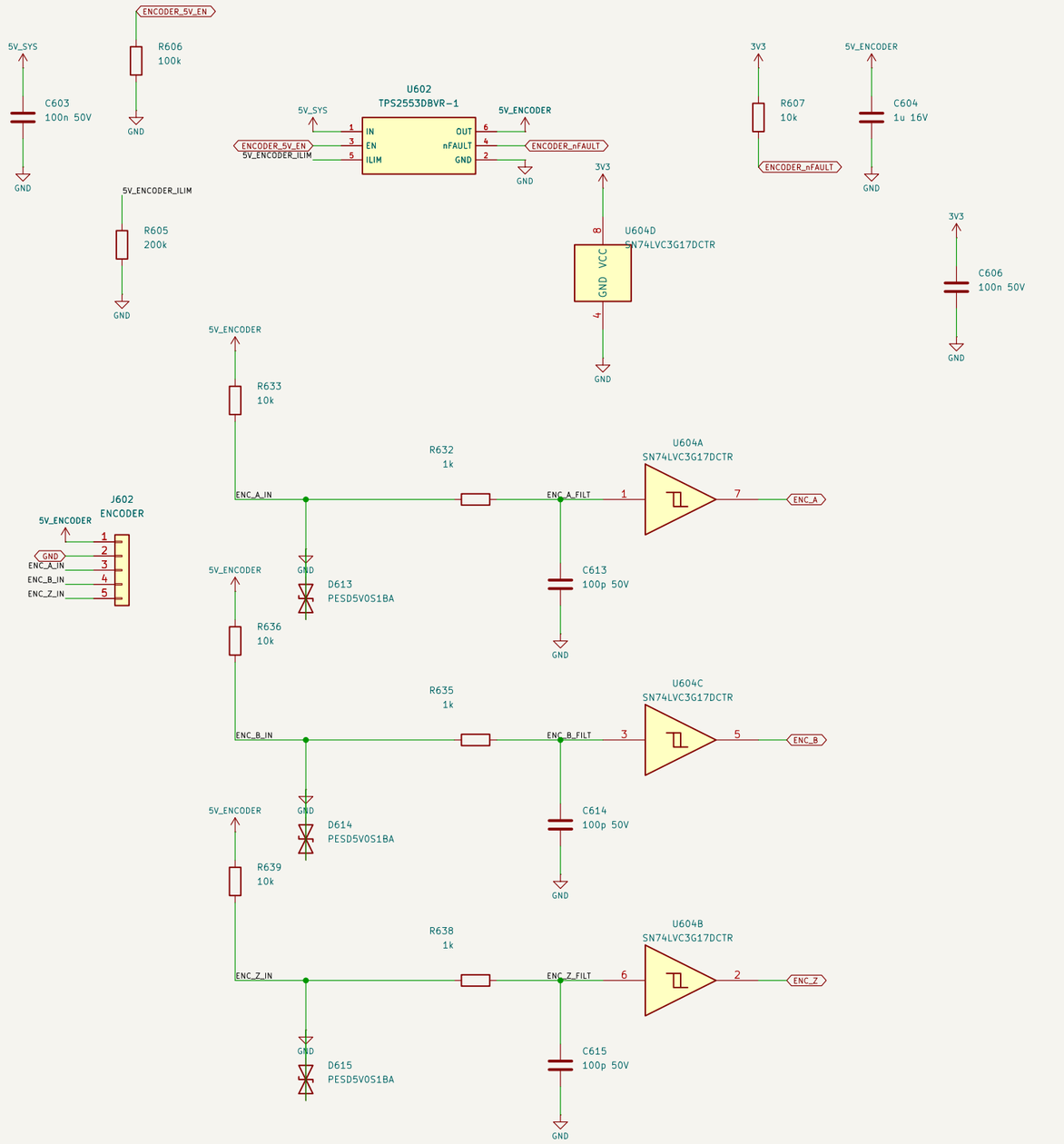
Hall interface



5 V supply: current limited, latch-off on fault.
Inputs: 3.3/5 V single-ended, not 24 V.

Connector labels define the controller pinout only. Verify the OEM motor Hall supply, ground and signal wires before connection.

Encoder interface

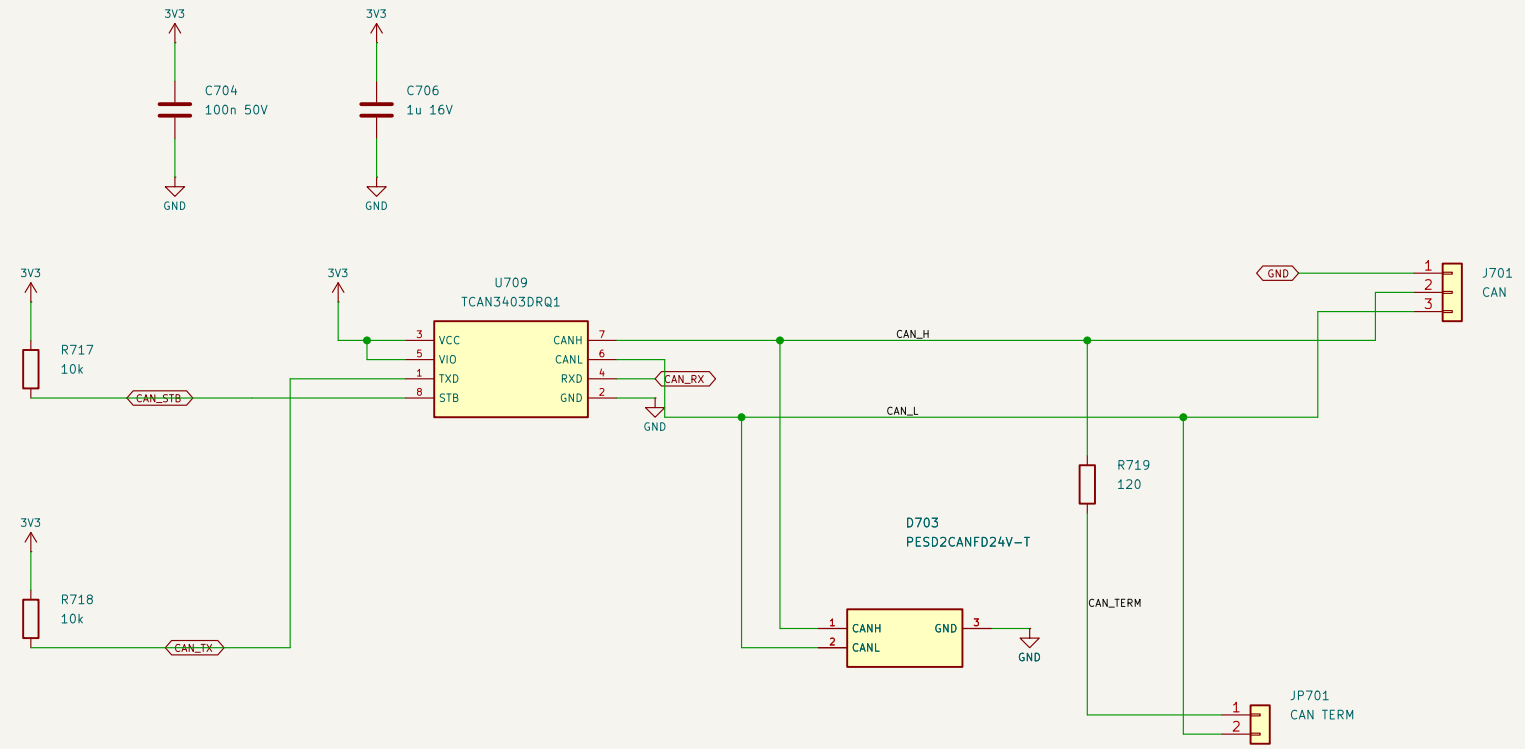


5 V supply: current limited, latch-off on fault.
Inputs: 3.3/5 V single-ended, not 24 V.

Sheet: /06_sensors/	
File: 06_sensors.kicad_sch	
Title: Hall and encoder interfaces	
Size: A3	Date:
KiCad E.D.A. 10.0.4	**Rev: v1-draft**
	Id: 7/17

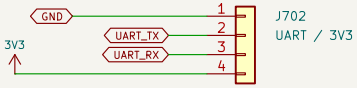
CAN FD and UART

CAN FD



JP701 fitted = 120 ohm termination. Fit termination only at the two ends of the CAN bus.

UART / 3.3 V logic

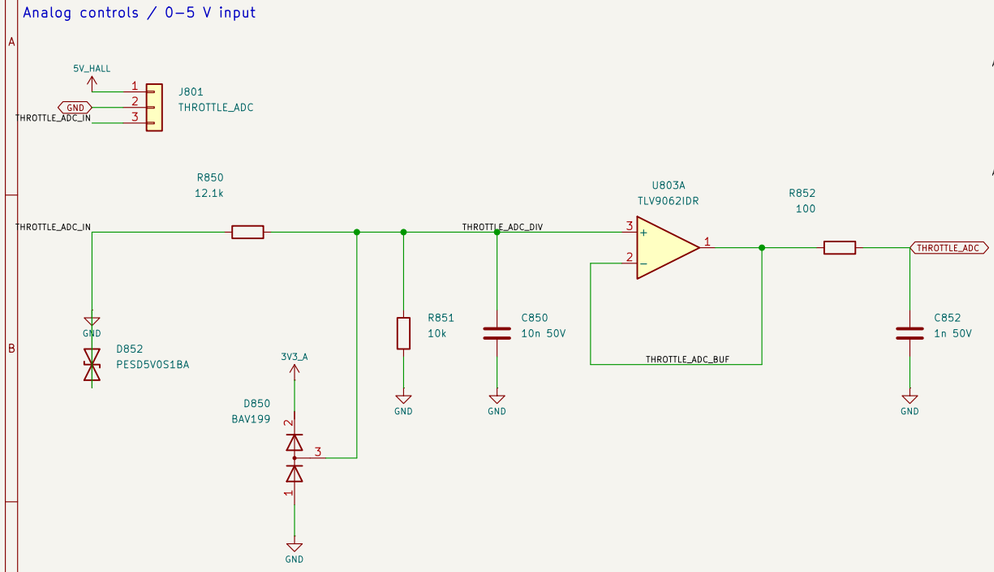


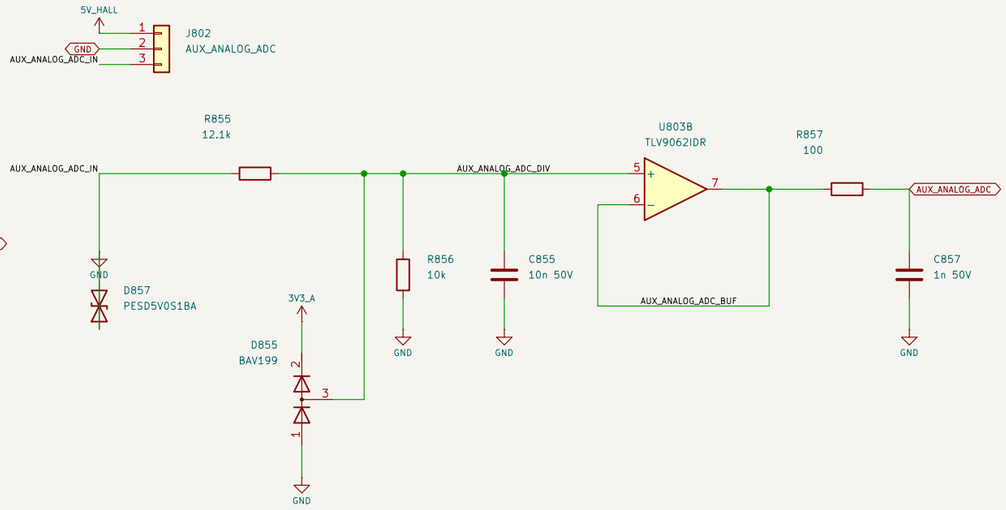
Short local 3.3 V wiring only: TX/RX are controller-side names.
Not an RS-232 or 5 V port. Supply pin is an output, not a power input.

Sheet: /07_io_can/		
File: 07_io_can.kicad_sch		
Title: CAN FD and UART		
Size: A4	Date:	**Rev: v1-draft**
KiCad E.D.A. 10.0.4		Id: 8/17

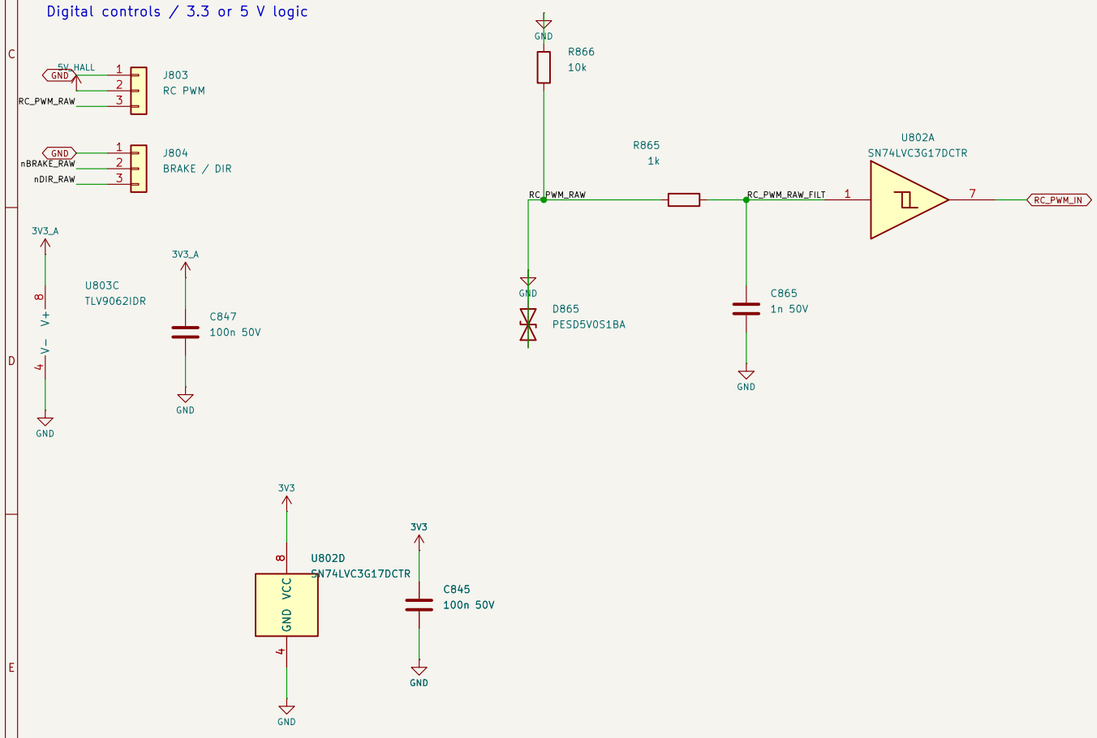
Analog and digital controls

Analog controls / 0–5 V input





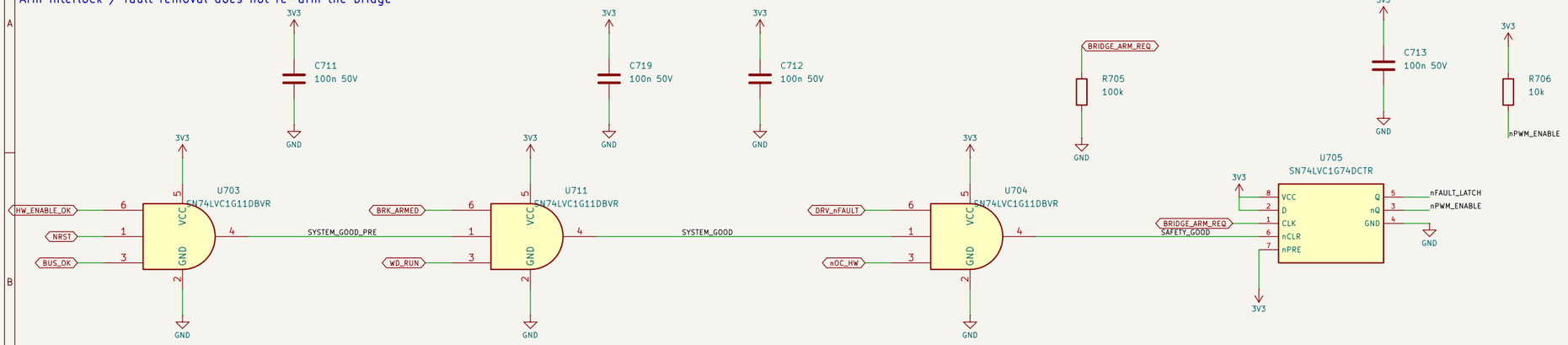
Digital controls / 3.3 or 5 V logic



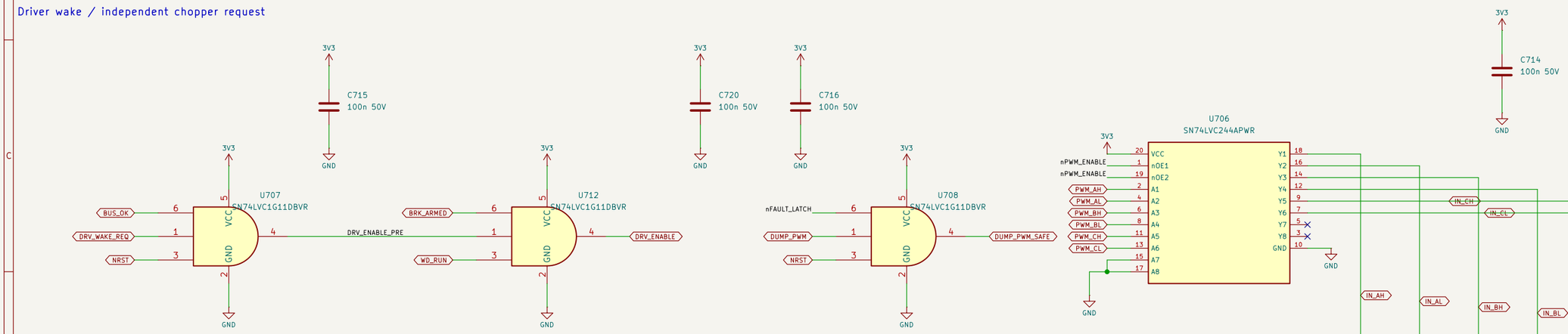
Sheet: /08_analog_inputs/		
File: 08_analog_inputs.kicad_sch		
Title: Analog and digital controls		
Size: A3	Date:	**Rev: v1–draft**
KiCad E.D.A. 10.0.4		Id: 9/17

Hardware bridge interlock

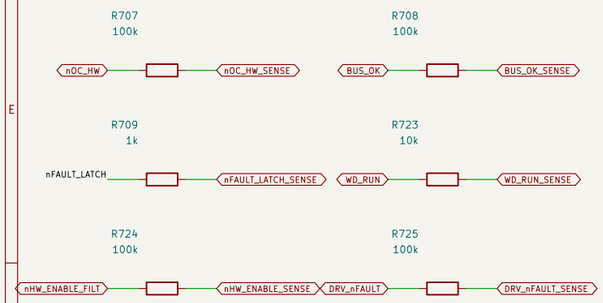
Arm interlock / fault removal does not re-arm the bridge



Driver wake / independent chopper request



MCU monitoring / series isolation

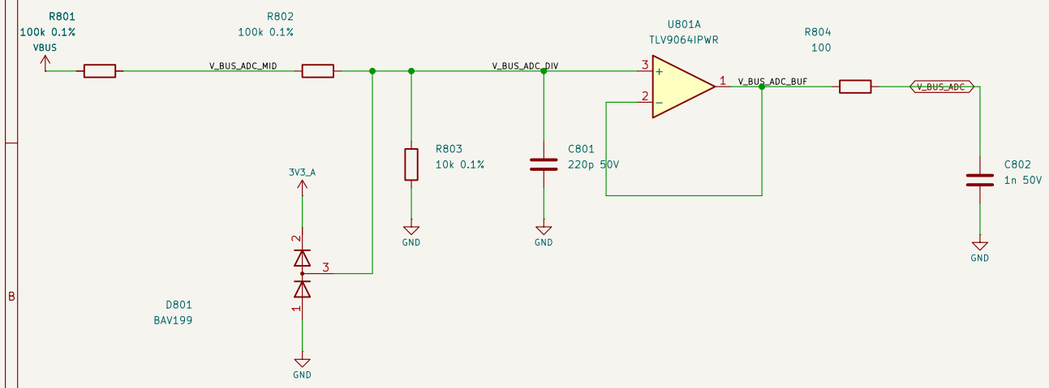


Hardware fault and enable nodes are isolated from MCU GPIO drive.
Disable UCPD dead-battery control before reading PB4.

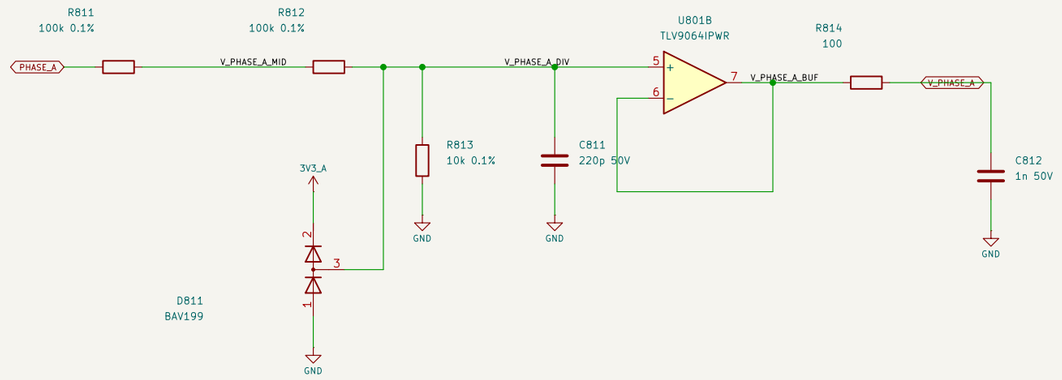
Sheet: /09_safety/	
File: 09_safety.kicad_sch	
Title: Hardware bridge interlock	
Size: A3	Date:
KiCad E.D.A. 10.0.4	Id: 10/17
	Rev: v1-draft

Bus and phase voltage sensing

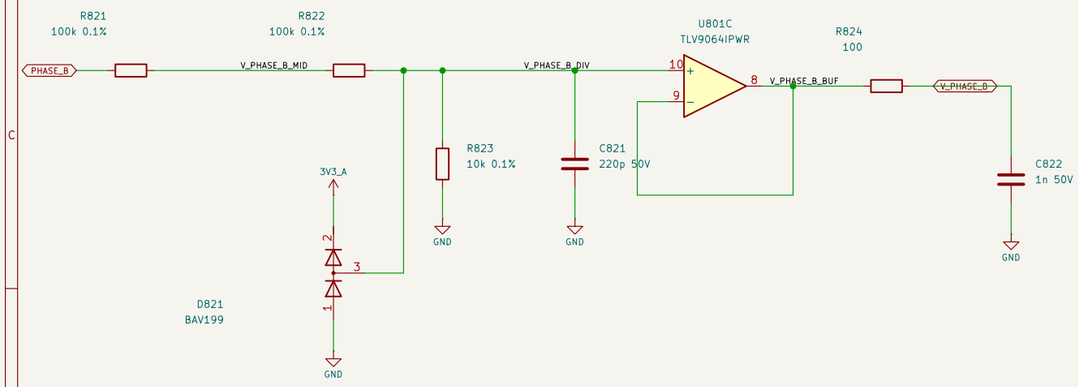
Bus voltage



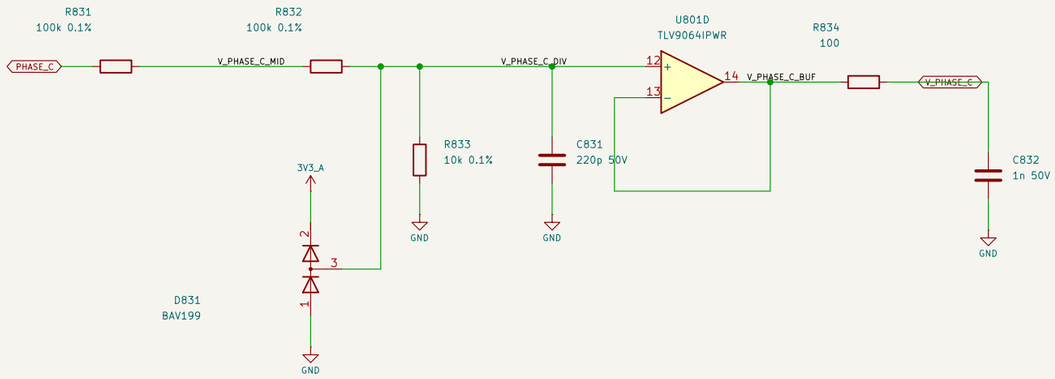
Phase A voltage

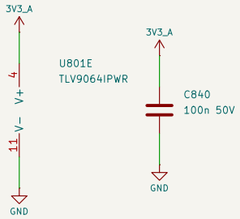


Phase B voltage



Phase C voltage





21:1 dividers: 2.5 V ADC full scale = 52.5 V at the input.
Buffers isolate the dividers from ADC sampling transients.
Normal operation remains limited to a 12–30 V bus.

Sheet: /10_voltage_sensing/		
File: 10_voltage_sensing.kicad_sch		
Title: Bus and phase voltage sensing		
Size: A3	Date:	**Rev: v1-draft**
KiCad E.D.A. 10.0.4		Id: 11/17

Current and bus protection

Bipolar phase−current window

Phase A / upper

Phase B / upper

Phase C / upper

U101A
TLV9024PWR

U101D
TLV9024PWR

U102A
TLV9024PWR

OC_HI

I_PHASE_A

I_PHASE_B

I_PHASE_C

Phase A / lower

Phase B / lower

Phase C / lower

U101B
TLV9024PWR

U101C
TLV9024PWR

U102B
TLV9024PWR

OC_LO

3V3

R116
4.7k

nOC_HW

Current thresholds / about +/−50 A

VREF_2V5

R110
10k 0.1%

R111
90.9k 0.1%

C110
10n 50V

R112
90.9k 0.1%

R113
10k 0.1%

C111
10n 50V

GND

Bus window / about 10.5−33 V

U102D
TLV9024PWR

U102C
TLV9024PWR

BUS_OV_REF

V_BUS_ADC

BUS_UV_REF

R117
10k

BUS_OK

R114
10k 0.1%

R115
16.9k 0.1%

C112
10n 50V

R118
40.2k 0.1%

R119
10k 0.1%

C117
10n 50V

3V3_A

U101E
TLV9024PWR

U102E
TLV9024PWR

C113
100n 50V

C114
100n 50V

Board protection only.
Use lower motor limits.
OEM motor: 0.75 A.

Sheet: /11_current_protection/
File: 11_current_protection.kicad_sch

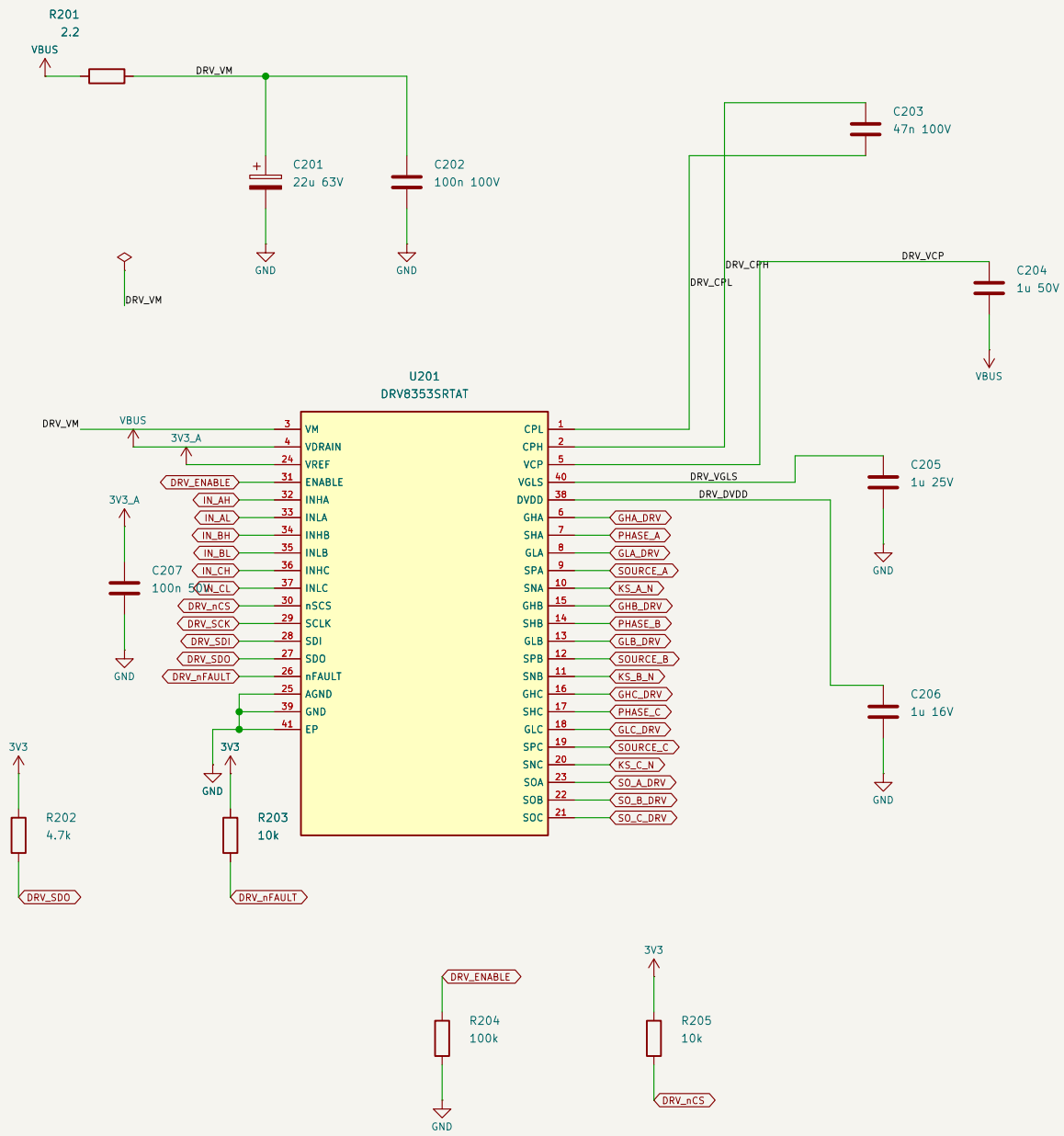
Title: Current and bus protection

Size: A4 | Date: | **Rev: v1−draft**

KiCad E.D.A. 10.0.4 | Id: 12/17

Gate driver and charge pump

Gate driver / charge pump



VM filter and pump capacitors belong next to U201. VCP reservoir returns to VBUS, not GND.

SPx returns directly to each low-side source. SNx / CSA inputs use Kelvin shunt sense pads.

Six PWM inputs arrive through the hardware arm interlock. ENABLE is separate for driver setup.

Sheet: /12_gate_driver/	
File: 12_gate_driver.kicad_sch	
Title: Gate driver and charge pump	
Size: A4	Date:
KiCad E.D.A. 10.0.4	**Rev: v1-draft**
	Id: 13/17

USB and status indicators

USB-C device port

F401
0.5A / 6V
USB_VBUS_RAW
USB_5V
C413
100n 50V
GND
J401
USB4105-GF-A
USB_VBUS_RAW
A4 VBUS
A9 VBUS
B4 VBUS
B9 VBUS
CC1 A5 CC1
CC2 B5 CC2
A8 SBU1
B8 SBU2
A6 D+ USB_DP_PORT
B6 D+ USB_DP_PORT
A7 D- USB_DM_PORT
B7 D- USB_DM_PORT
A1 GND
A12 GND
B1 GND
B12 GND
SH SHIELD CHASSIS
GND
CC1
R411
5.1k
GND
CC2
R412
5.1k
GND
5V_BUS
USB_VBUS_RAW
USB_5V
U404
USBLC6-2SC6
USB_VBUS_RAW
USB_DP_PORT
USB_DM_PORT
R413
22
USB_DP_ESD
USB_DP
R414
22
USB_DM_ESD
USB_DM
GND
R415
10k
USB_VBUS_SENSE
R416
15k
GND

Status / probe points

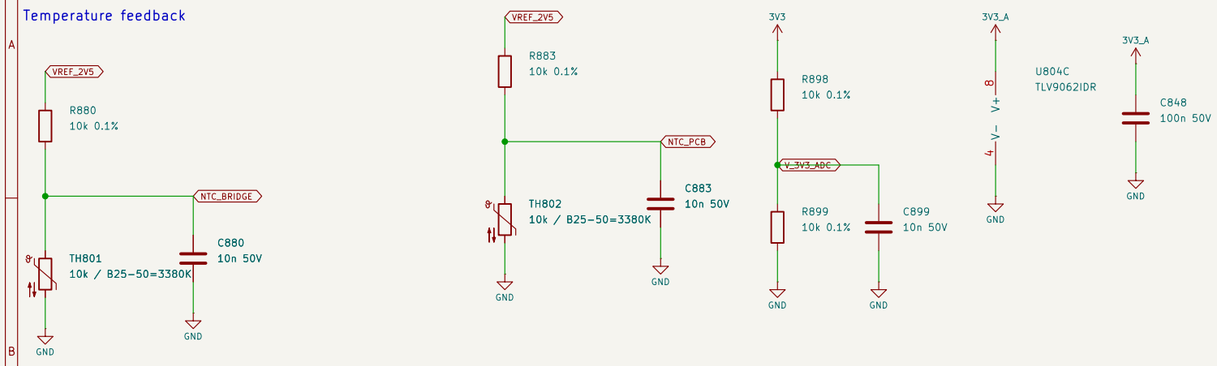
LED_STATUS
R511
1k
LED_1
D511
STATUS
GND
LED_COMM
R512
1k
LED_2
D512
COMM
GND
3V3
R513
1k
LED_3
D513
POWER
GND
TP401
5V_BUS
5V_BUS
TP402
5V_SYS
5V_SYS
TP403
3V3
3V3
TP404
GND
GND

USB shield bonds to CHASSIS. D+/D- protection belongs close to the connector; 22 ohm resistors belong near the MCU.

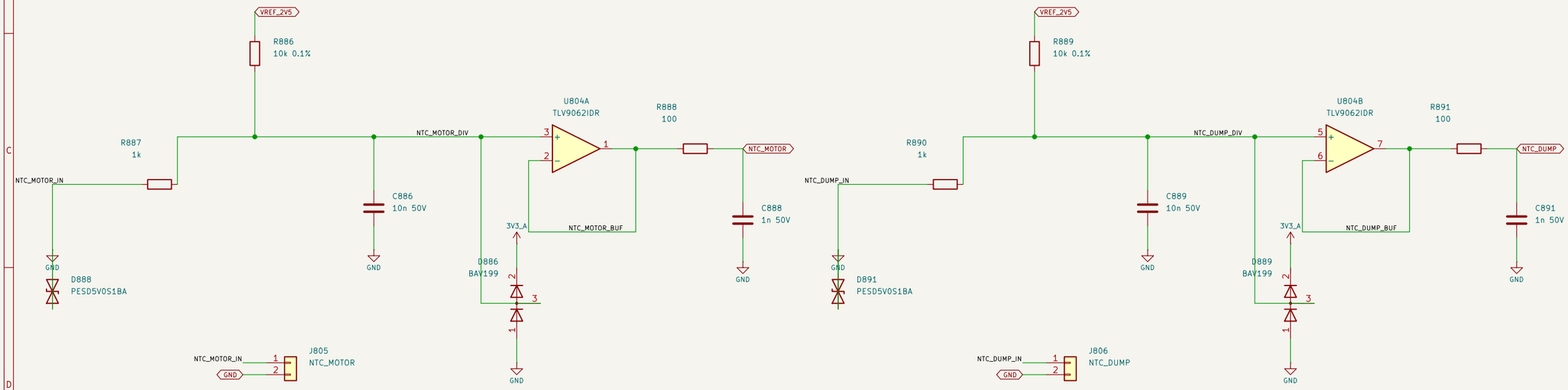
Sheet:	/13_usb_status/
File:	13_usb_status.kicad_sch
Title:	USB and status indicators
Size:	A3
Date:	
Rev:	v1-draft
KiCad E.D.A.	10.0.4
Id:	14/17

Temperature and logic-rail feedback

Temperature feedback



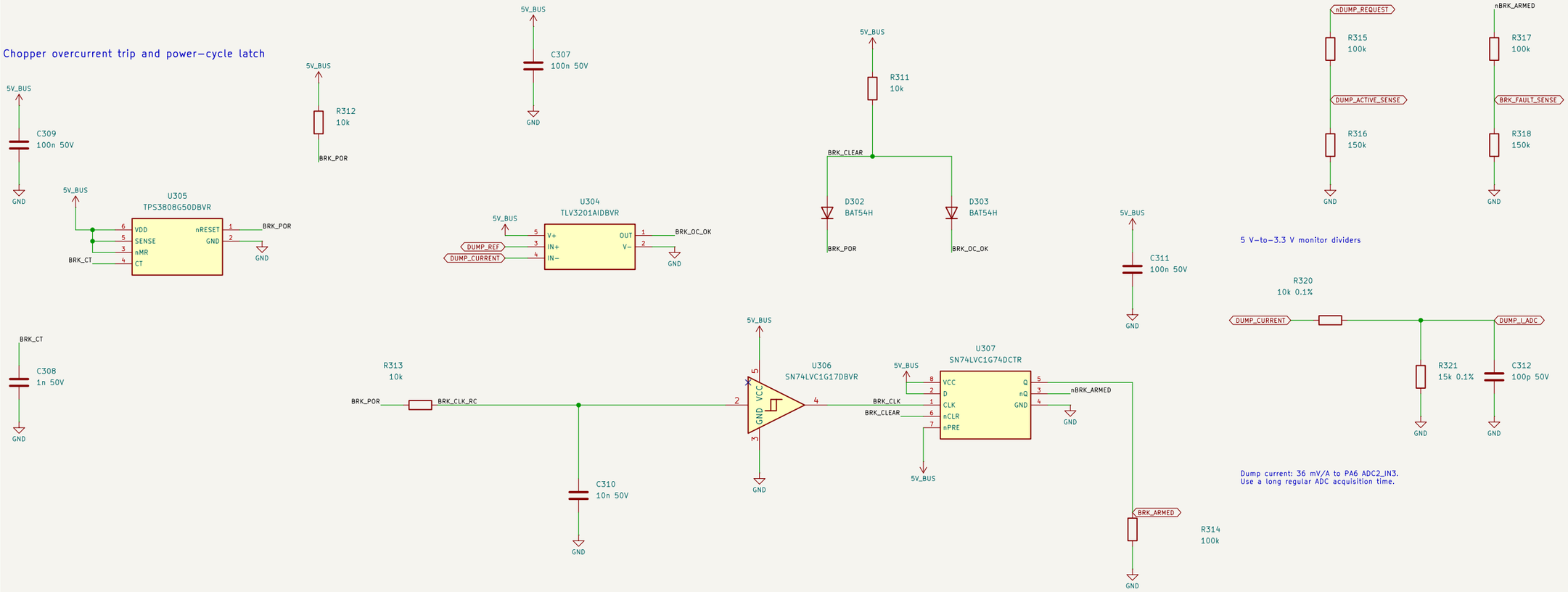
TH801/TH802: 10k, B25/50=3380 K. External NTC conversion must subtract the 1k series resistor.
External channels are buffered; configure their actual R/T curves before thermal-limit testing.



Sheet: /14_temperature/		
File: 14_temperature.kicad_sch		
Title: Temperature and logic-rail feedback		
Size: A3	Date:	**Rev: v1-draft**
KiCad E.D.A. 10.0.4		Id: 15/17

Brake protection and telemetry

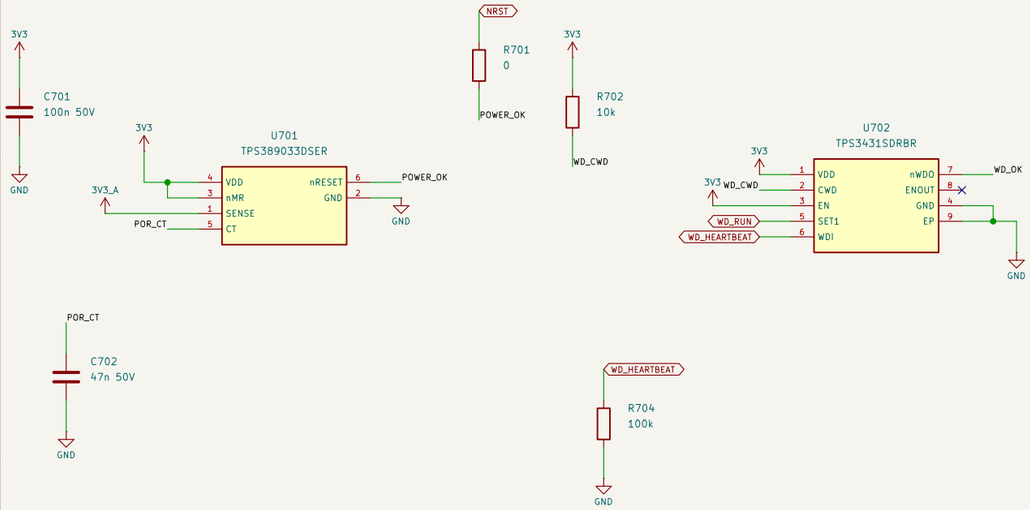
Chopper overcurrent trip and power-cycle latch



Sheet: /15_regen_safety/		
File: 15_regen_safety.kicad_sch		
Title: Brake protection and telemetry		
Size: A3	Date:	**Rev: v1-draft**
KiCad E.D.A. 10.0.4		Id: 16/17

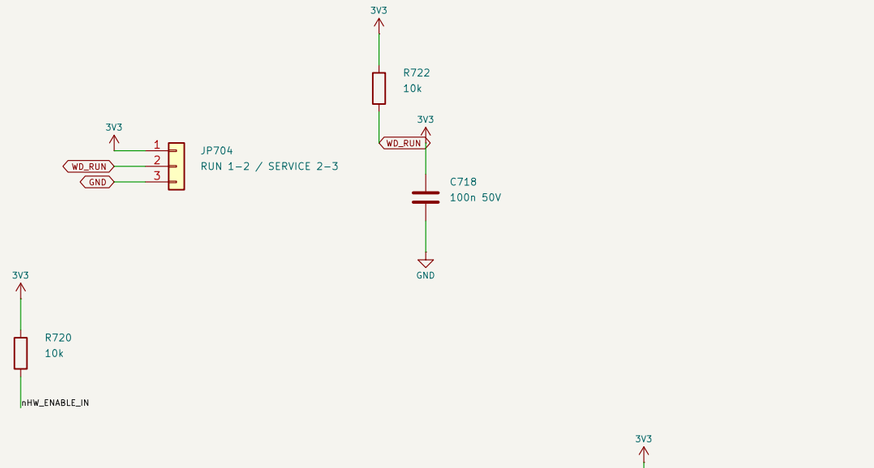
Reset, watchdog and hardware enable

Power-on reset and watchdog



Open-drain reset outputs share NRST through zero-ohm links.
R504 on the MCU sheet provides the pull-up.

Service and hardware enable



RUN 1-2 / SERVICE 2-3.
SERVICE or a chopper fault forces
both driver-enable and bridge-arm paths off.

Sheet: /16_supervisor/		
File: 16_supervisor.kicad_sch		
Title: Reset, watchdog and hardware enable		
Size: A3	Date:	**Rev: v1-draft**
KiCad E.D.A. 10.0.4		Id: 17/17